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Sunohara et al.

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(54) **SUBSTRATE FIXING DEVICE**

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H01L 21/687 (2006.01)

(52) **U.S. Cl.**

CPC **H01L 21/6833** (2013.01); **H01L 21/68757**
(2013.01)

(58) **Field of Classification Search**

CPC H01L 21/6833; H01L 21/68757

See application file for complete search history.

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(57) **ABSTRACT**

A substrate fixing device includes a base plate, a ceramic plate fixed to the base plate and configured to adsorb a substrate by electrostatic force, and an adhesive layer bonding the base plate and the ceramic plate. The adhesive layer includes a plurality of linear heat transfer bodies arranged adjacent to each other such that a longitudinal direction of each linear heat transfer body is along a stack direction of the ceramic plate and the base plate, and a resin filled between adjacent heat transfer bodies of the heat transfer bodies and bonded to the ceramic plate and the base plate.

5 Claims, 17 Drawing Sheets

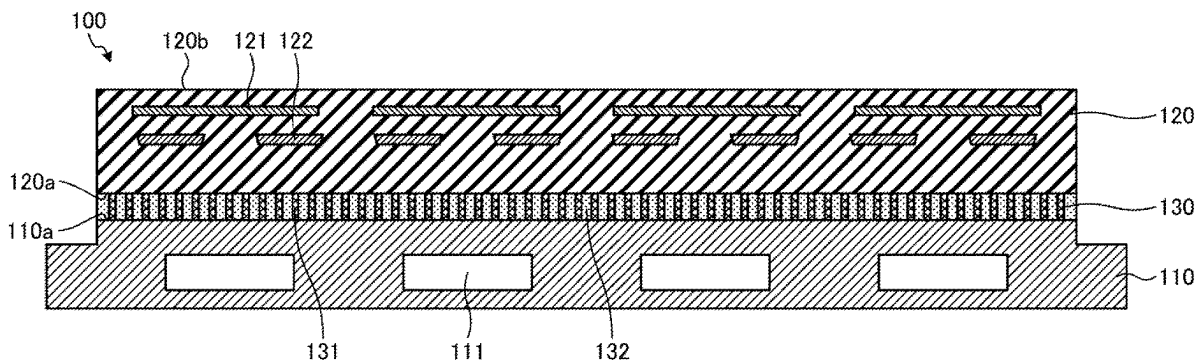


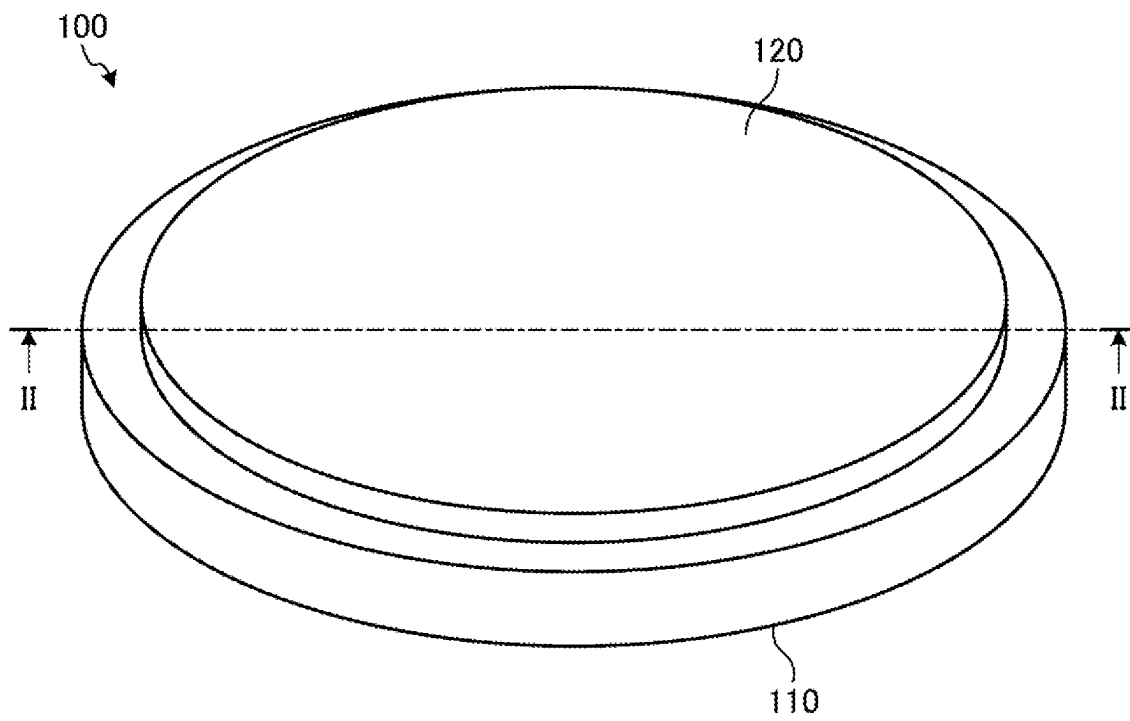
FIG. 1

FIG. 2

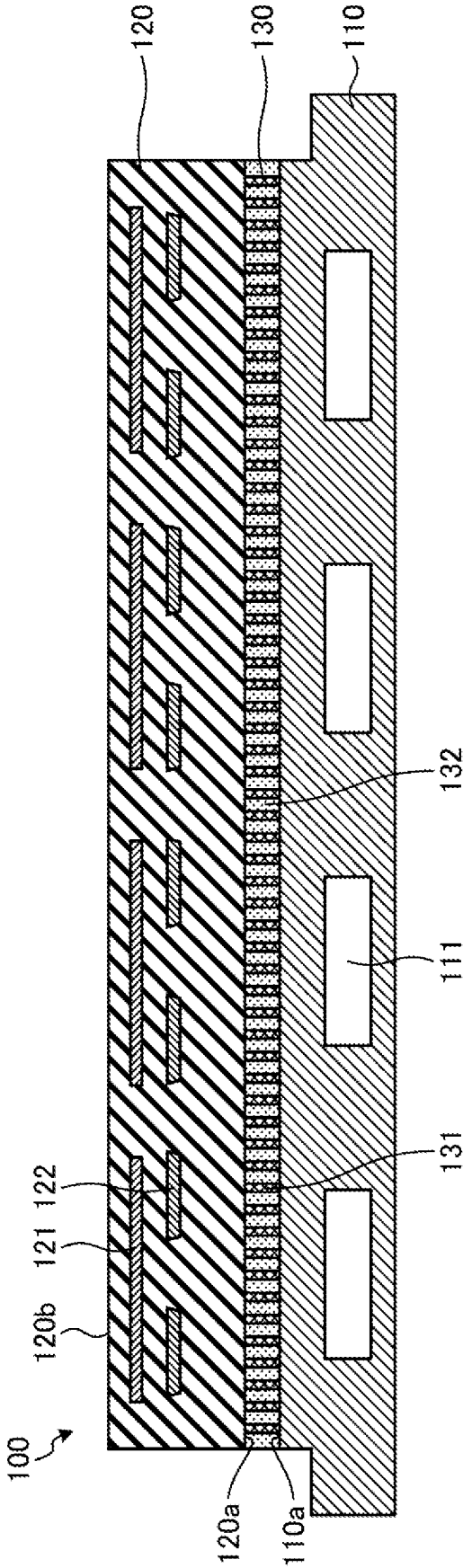


FIG. 3

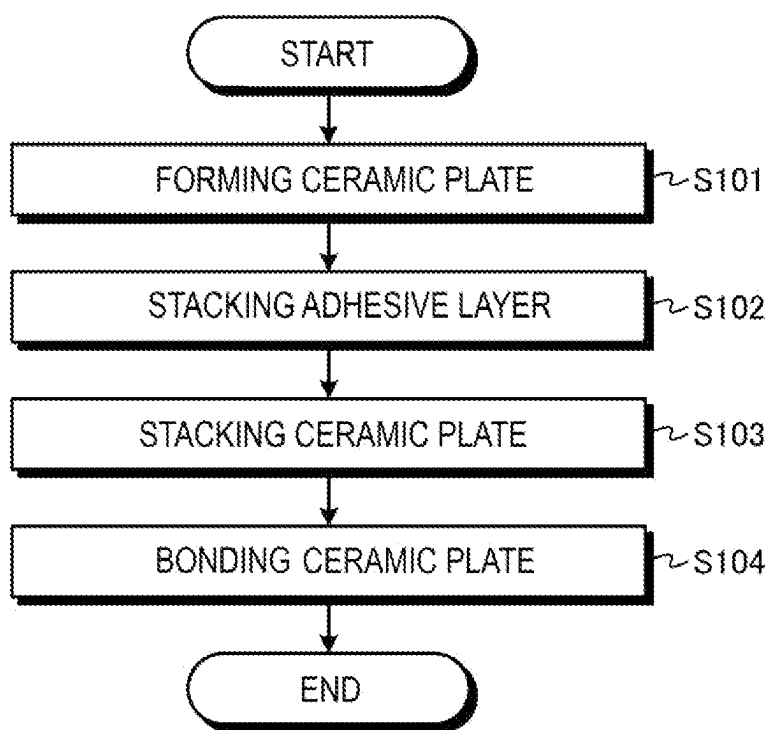


FIG. 4

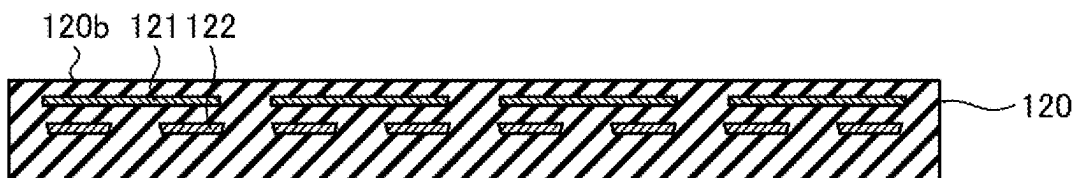


FIG. 5

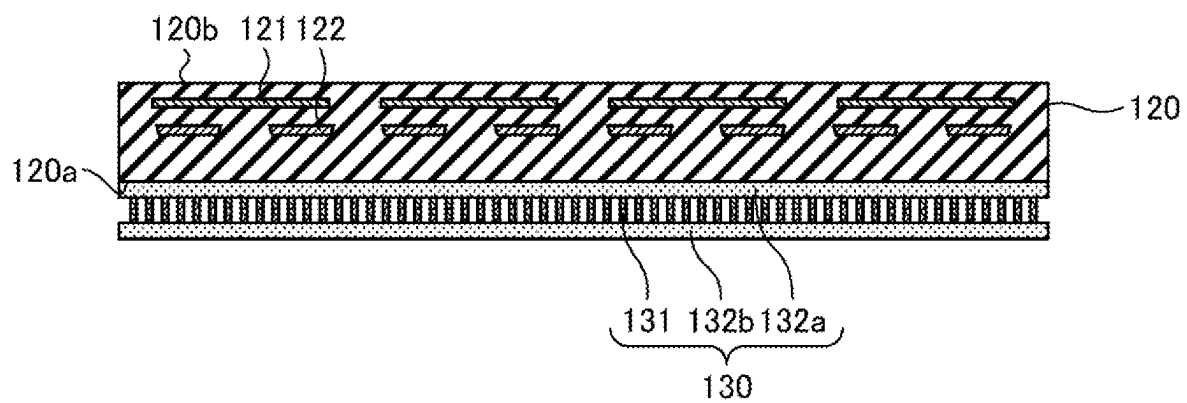


FIG. 6

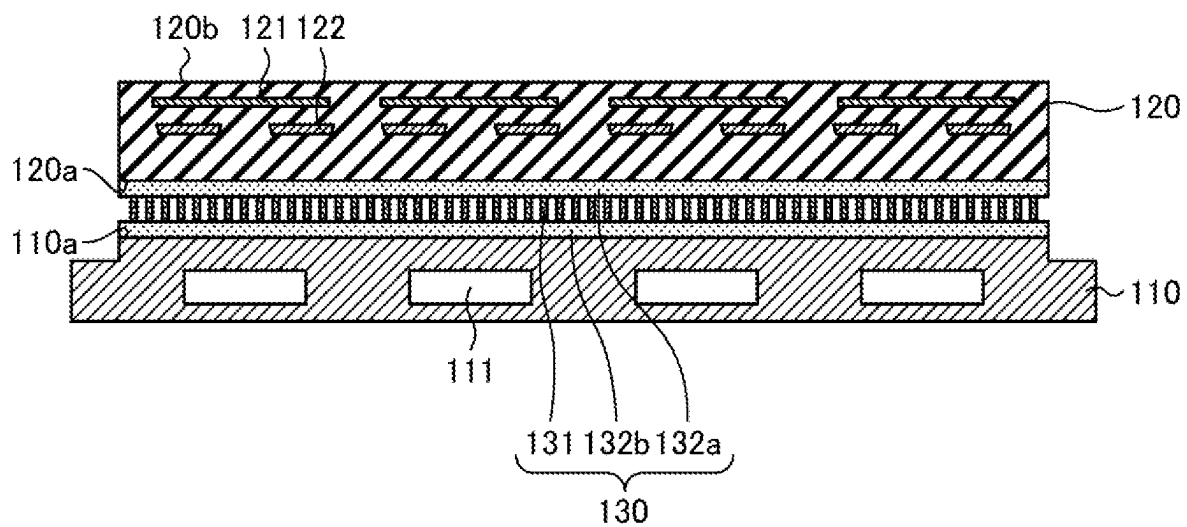


FIG. 7

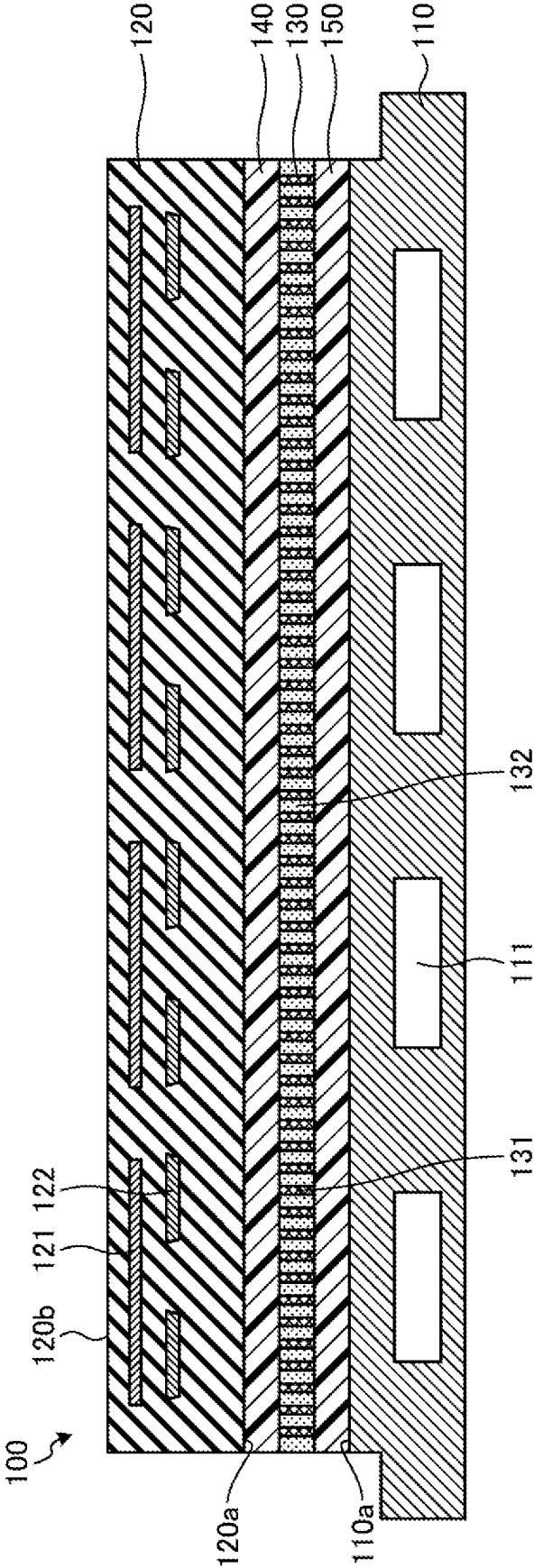


FIG. 8

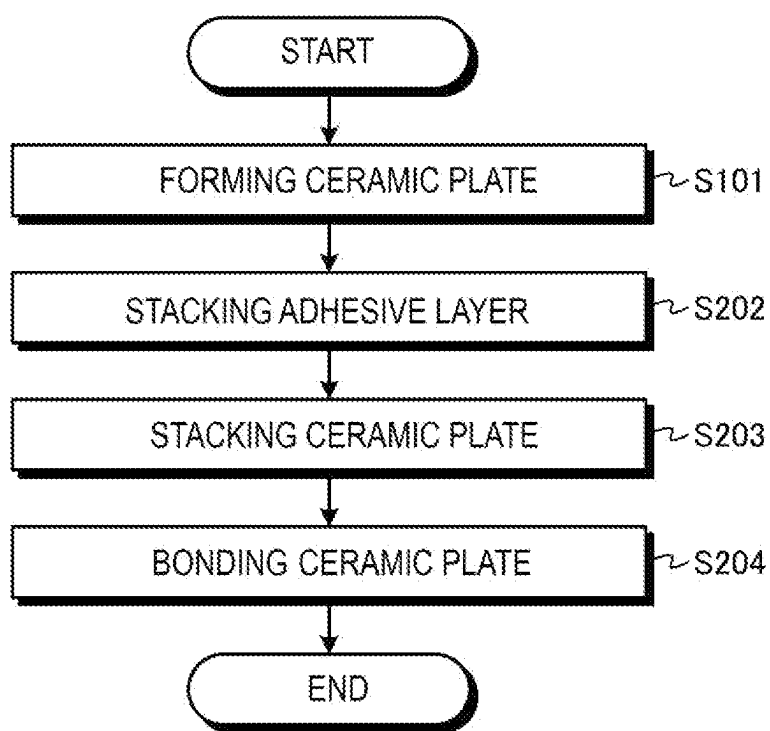


FIG. 9

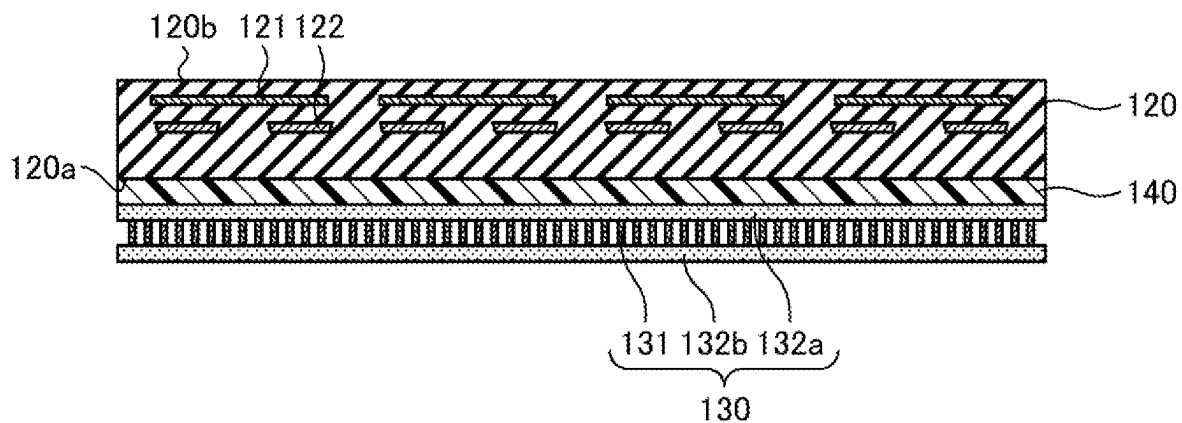


FIG. 10

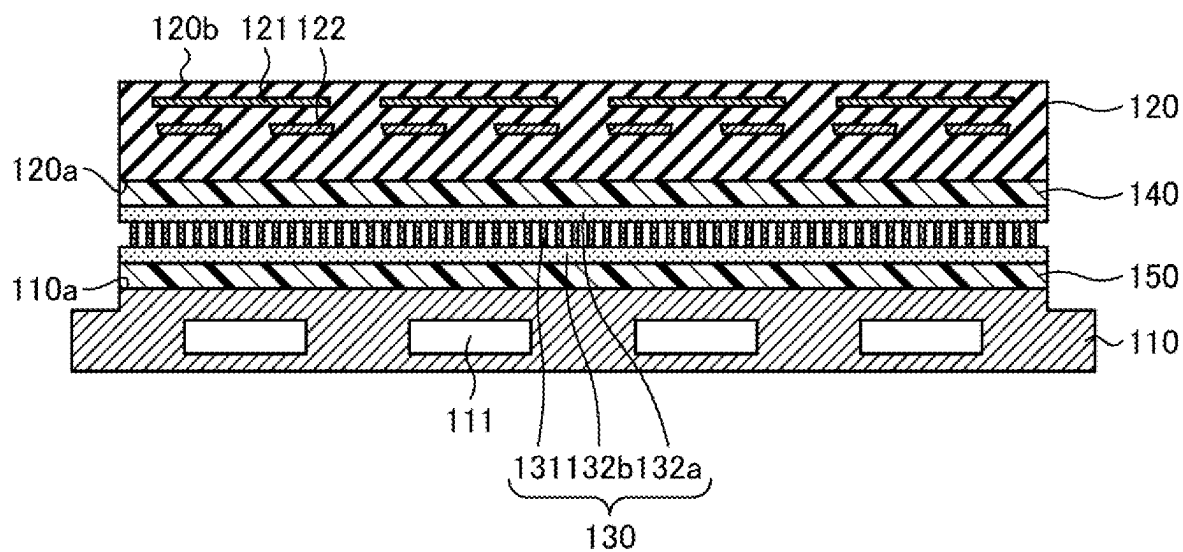


FIG. 11

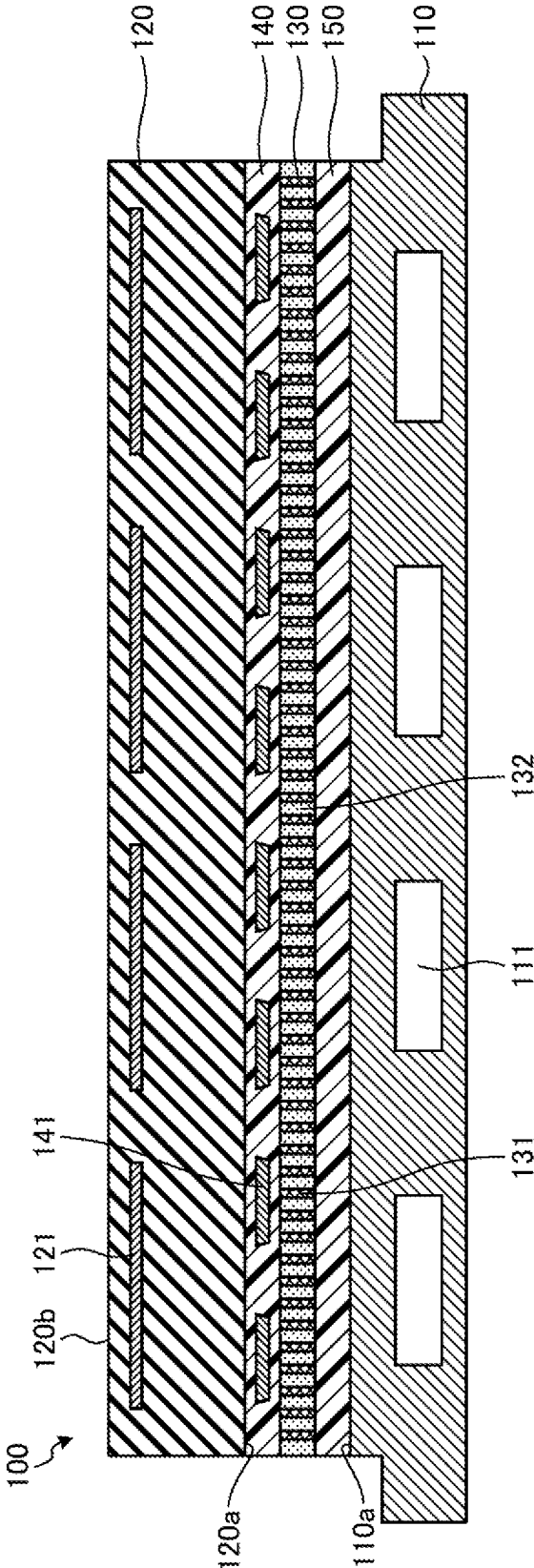


FIG. 12

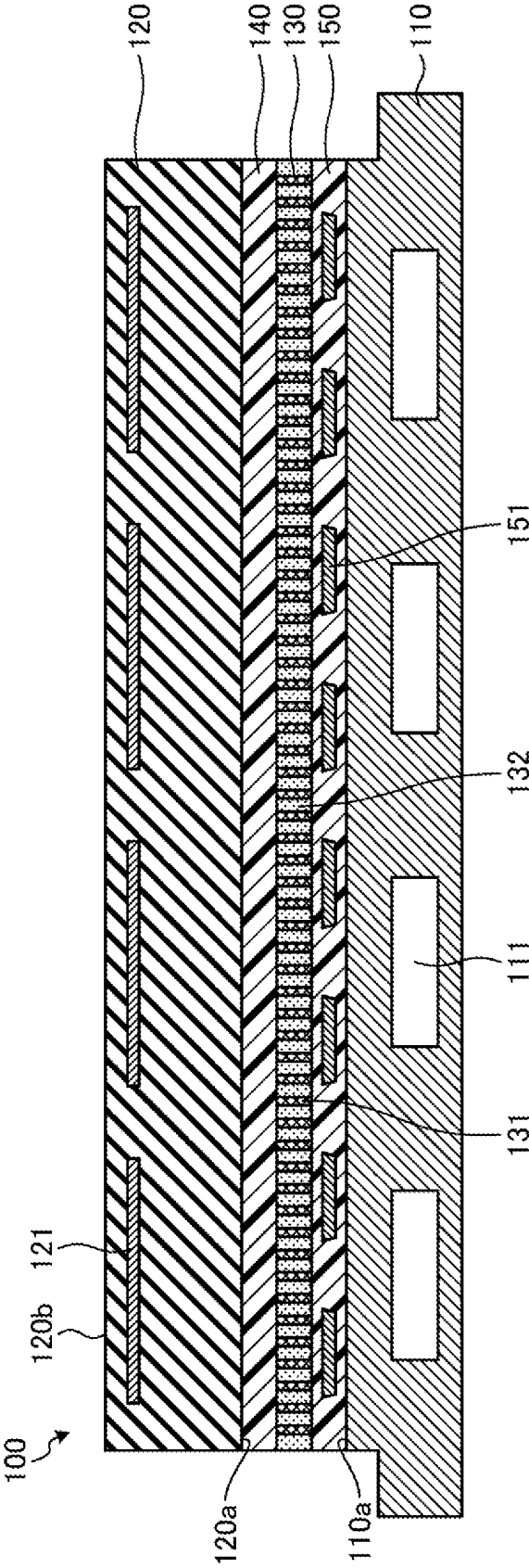


FIG. 13

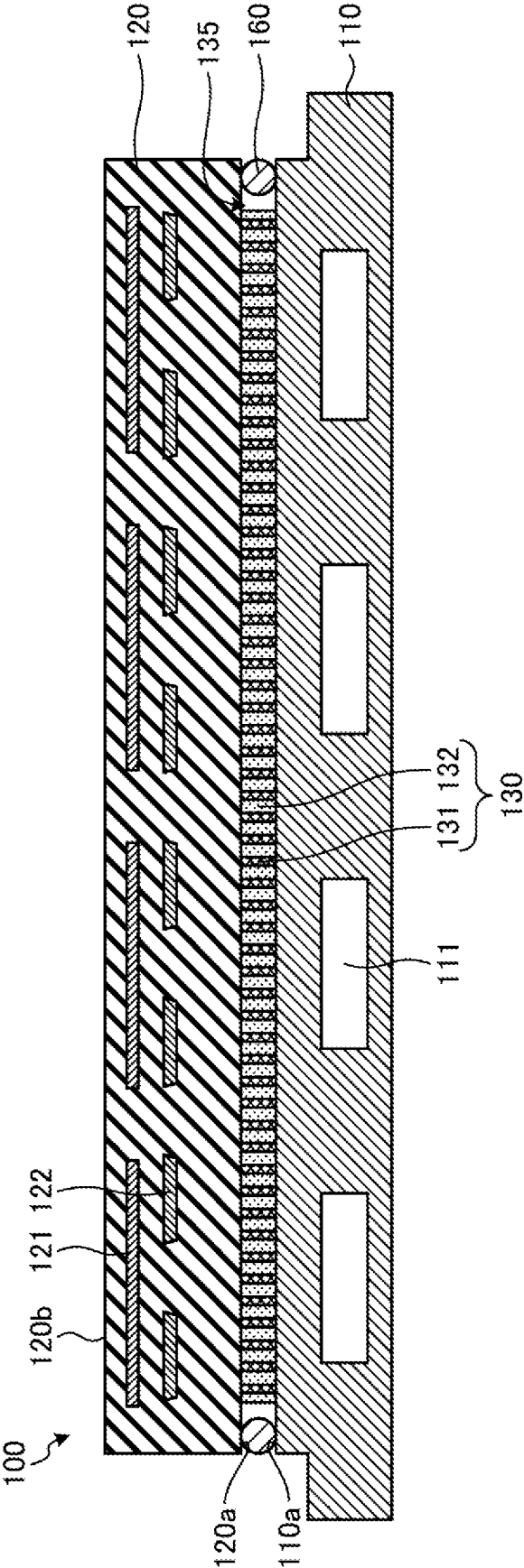


FIG. 14

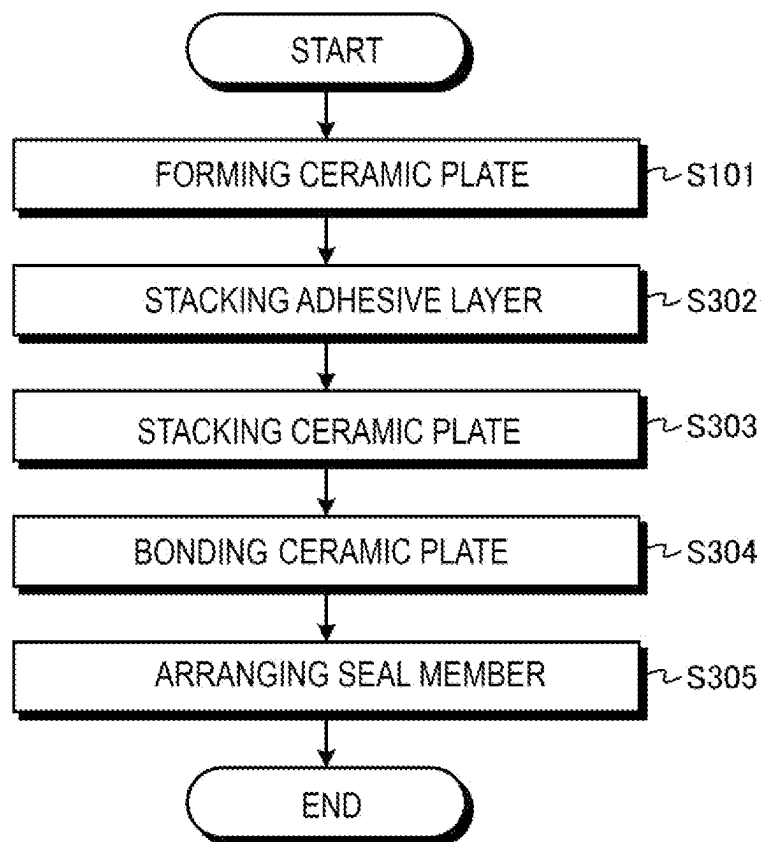


FIG. 15

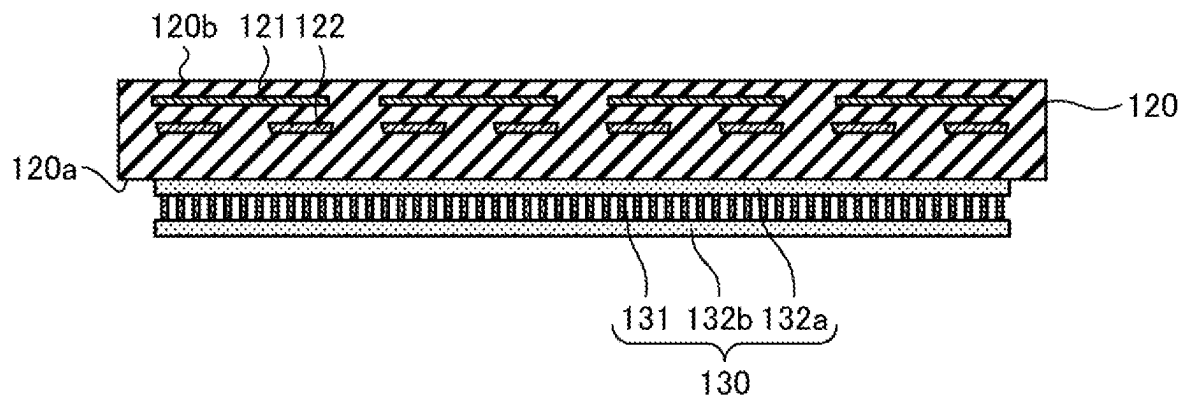


FIG. 16

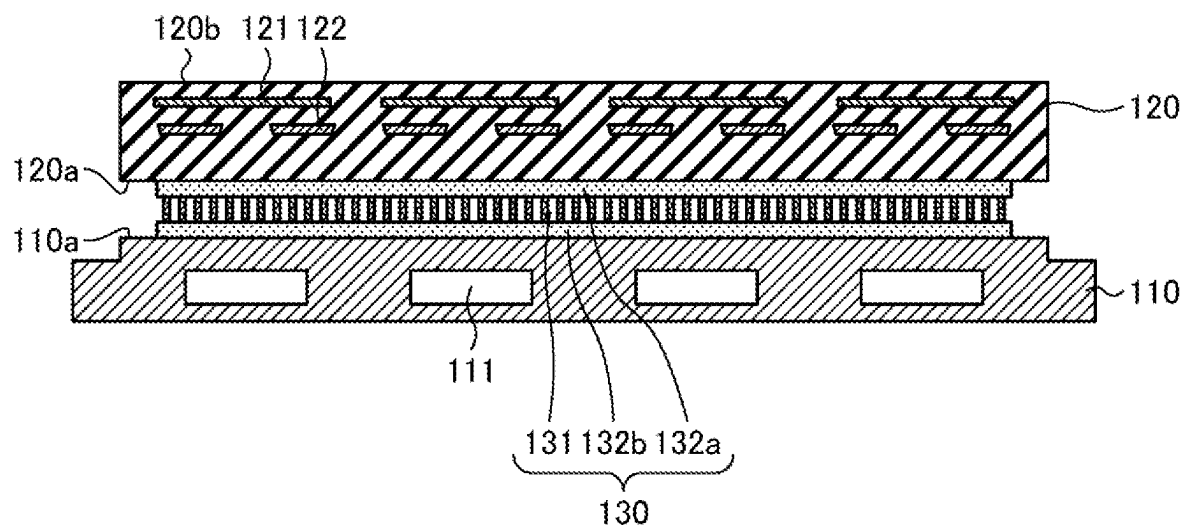


FIG. 17

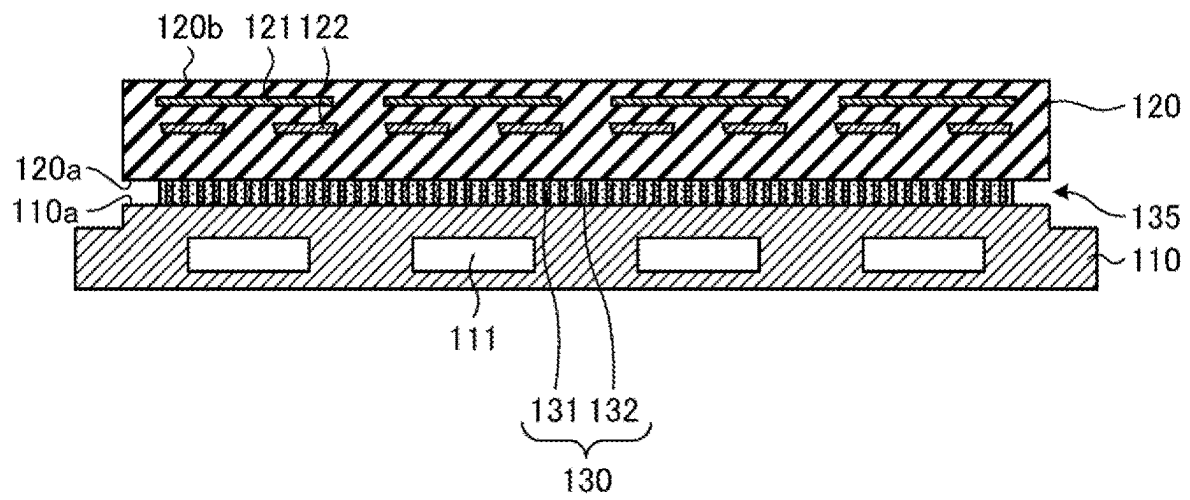


FIG. 19

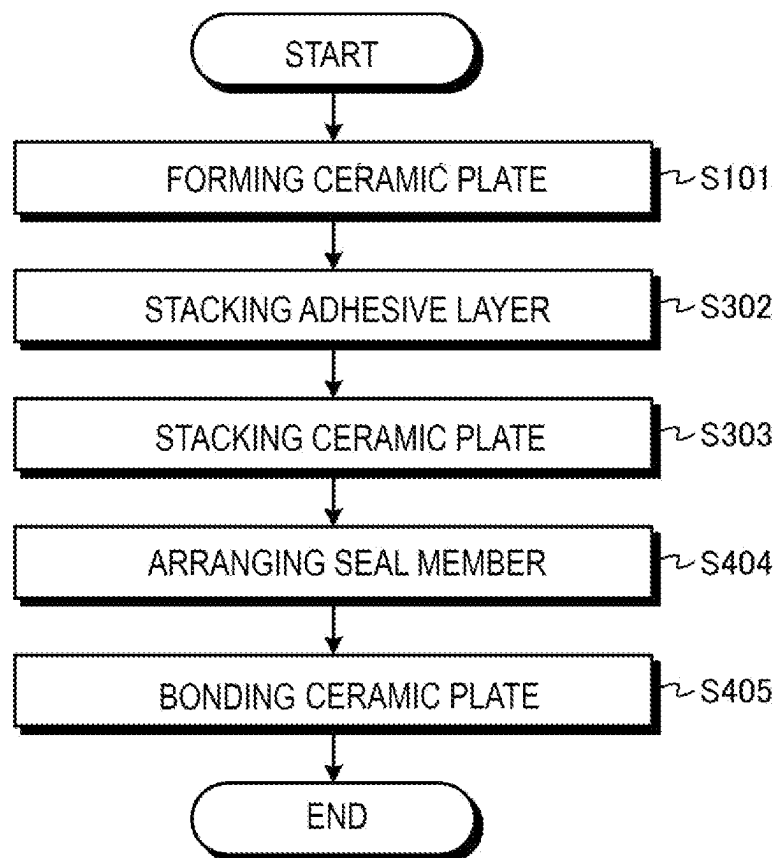
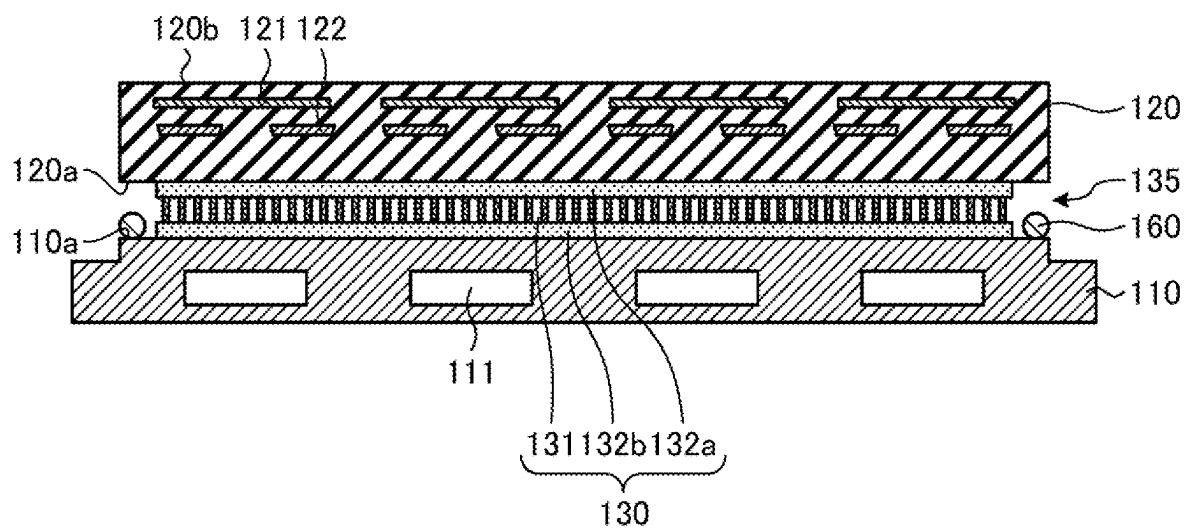


FIG. 20



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SUBSTRATE FIXING DEVICE**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is based on and claims the benefit of priority from prior Japanese patent application No. 2022-097236 filed on Jun. 16, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

The present invention relates to a substrate fixing device.

BACKGROUND ART

In general, a substrate fixing device that adsorbs and holds a wafer when manufacturing a semiconductor component, for example, is also referred to as an electrostatic chuck (ESC), and includes a ceramic plate having an electrode embedded therein. The substrate fixing device has a structure in which the ceramic plate is bonded to a base plate, and applies a voltage to the electrode embedded in the ceramic plate, thereby adsorbing the wafer on the ceramic plate by using electrostatic force. By adsorbing and holding the wafer on the ceramic plate, processes such as microfabrication and etching on the wafer are efficiently performed.

In such a substrate fixing device, the ceramic plate is bonded to the base plate by, for example, a silicone resin-based adhesive. When the ceramic plate and the base plate are bonded by an adhesive, since the thermal resistance of the adhesive in a thickness direction is relatively large, the transfer of heat from the ceramic plate adsorbing the wafer to the base plate is inhibited, which may make it difficult to rapidly control a temperature of the wafer. Regarding this, in order to improve the heat transfer property from the ceramic plate to the base plate, a technology of bonding the ceramic plate to the base plate by an adhesive layer composed of a carbon nanotube assembly having high thermal conductivity in a longitudinal direction, instead of the adhesive, is suggested.

CITATION LIST**Patent Literature**

PTL 1: JP2021-111688A

SUMMARY OF INVENTION

However, when the adhesive layer composed of the carbon nanotube assembly is used, the thermal responsiveness in the adhesive layer is improved, but there is a problem in that the adhesiveness may decrease. Specifically, when the ceramic plate and the base plate are bonded based only on the intermolecular force of the carbon nanotubes in the adhesive layer, the bonding strength of the ceramic plate and the base plate is relatively low. For this reason, the adhesive layer may peel off from the ceramic plate or the base plate due to the stress caused by a difference in thermal expansion coefficient between the ceramic plate and the base plate.

Aspect of non-limiting embodiments of the present disclosure is to provide a substrate fixing device capable of improving thermal responsiveness and adhesiveness in an adhesive layer.

According to an aspect of the present disclosure, there is provided a substrate fixing device including a base plate, a

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ceramic plate, and an adhesive layer. The ceramic plate is fixed to the base plate and is configured to adsorb a substrate by electrostatic force. The adhesive layer is configured to bond the base plate and the ceramic plate. The adhesive layer has a plurality of linear heat transfer bodies and a resin. The plurality of heat transfer bodies are arranged adjacent to each other so that a longitudinal direction of each linear heat transfer body is along a stack direction of the ceramic plate and the base plate. The resin is filled between adjacent heat transfer bodies of the heat transfer bodies and is bonded to the ceramic plate and the base plate.

According to one aspect of the substrate fixing device, it is possible to achieve the effect of improving thermal responsiveness and adhesiveness in the adhesive layer.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a perspective view showing a configuration of a substrate fixing device according to a first embodiment.

FIG. 2 is a schematic view showing a cross section of the substrate fixing device according to the first embodiment.

FIG. 3 is a flowchart of showing a manufacturing method of the substrate fixing device according to the first embodiment.

FIG. 4 shows a specific example of a ceramic plate.

FIG. 5 shows a specific example of an adhesive layer stacking process.

FIG. 6 shows a specific example of a ceramic plate stacking process.

FIG. 7 is a schematic view showing a cross section of a substrate fixing device according to a second embodiment.

FIG. 8 is a flowchart showing a manufacturing method of the substrate fixing device according to the second embodiment.

FIG. 9 shows a specific example of an adhesive layer stacking process.

FIG. 10 shows a specific example of a ceramic plate stacking process.

FIG. 11 shows a first modified embodiment of the substrate fixing device according to the second embodiment.

FIG. 12 shows a second modified embodiment of the substrate fixing device according to the second embodiment.

FIG. 13 is a schematic view showing a cross section of a substrate fixing device according to a third embodiment.

FIG. 14 is a flowchart showing a manufacturing method of the substrate fixing device according of the third embodiment.

FIG. 15 shows a specific example of an adhesive layer stacking process.

FIG. 16 shows a specific example of a ceramic plate stacking process.

FIG. 17 shows a specific example of an intermediate structure.

FIG. 18 is a schematic view showing a cross section of a substrate fixing device according to a fourth embodiment.

FIG. 19 is a flowchart showing a manufacturing method of the substrate fixing device according of the fourth embodiment.

FIG. 20 shows a specific example of a seal member arranging process.

DESCRIPTION OF EMBODIMENTS

Hereinafter, embodiments of a substrate fixing device and a manufacturing method of a substrate fixing device disclosed by the present disclosure will be described in detail

with reference to the drawings. Note that the disclosed technology is not limited to the embodiments.

First Embodiment

FIG. 1 is a perspective view showing a configuration of a substrate fixing device 100 according to a first embodiment. The substrate fixing device 100 shown in FIG. 1 has a structure in which a ceramic plate 120 is bonded to a base plate 110.

The base plate 110 is a circular member made of metal such as aluminum, for example. The base plate 110 is a base member for fixing the ceramic plate 120. The base plate 110 is attached to a semiconductor manufacturing apparatus, for example, and causes the substrate fixing device 100 to function as a semiconductor holding device for holding a wafer.

The ceramic plate 120 is a circular member made of insulating ceramic. A diameter of the ceramic plate 120 is smaller than a diameter of the base plate 110, and the ceramic plate 120 is fixed to a center of the base plate 110. That is, a lower surface of the ceramic plate 120 serves as an adhesive surface that is bonded to the base plate 110, and the adhesive surface is bonded to the base plate 110, so that the ceramic plate 120 is fixed. An upper surface of the ceramic plate 120 is, for example, an adsorption surface for adsorbing a target object to be adsorbed, such as a wafer.

The ceramic plate 120 has an electrically conductive electrode embedded therein, and adsorbs a target object such as a wafer to the adsorption surface by using electrostatic force that is generated when a voltage is applied to the electrode. In addition, the ceramic plate 120 has a heater electrode embedded therein, and adjusts temperatures of the ceramic plate 120 and the target object such as a wafer adsorbed on the ceramic plate 120 by the heater electrode that generates heat when a voltage is applied thereto.

FIG. 2 is a schematic view showing a cross section of the substrate fixing device 100 according to the first embodiment. In FIG. 2, a cross section taken along a line II-II in FIG. 1 is shown. As shown in FIG. 2, the substrate fixing device 100 is configured by bonding the base plate 110 and the ceramic plate 120 with an adhesive layer 130.

The base plate 110 is a circular member made of metal and having a thickness of about 20 to 50 mm, for example. In the base plate 110, a refrigerant passage 111 serving as a passage for refrigerant such as cooling water or cooling gas is formed. The refrigerant passes through the refrigerant passage 111, so that the ceramic plate 120 is cooled. The ceramic plate 120 is cooled, so that the target object such as a wafer adsorbed on the adsorption surface of the ceramic plate 120 is cooled. An upper surface 110a of the base plate 110 is an adhesive surface that is bonded to the ceramic plate 120, and is bonded to a lower surface 120a of the ceramic plate 120 by the adhesive layer 130.

The ceramic plate 120 is a circular plate made of ceramic and having a thickness of 4 to 6 mm, for example. The ceramic plate 120 can be obtained by firing a green sheet fabricated using aluminum oxide, for example. The lower surface 120a of the ceramic plate 120 is an adhesive surface that is bonded to the base plate 110, and is bonded to the upper surface 110a of the base plate 110 by the adhesive layer 130. In the ceramic plate 120, an electrode 121 and a heater electrode 122 are formed.

The electrode 121 is arranged in the ceramic plate 120, and generates electrostatic force when a voltage is applied thereto. By the electrostatic force, the ceramic plate 120

adsorbs a target object such as a wafer to an upper surface 120b serving as an adsorption surface.

The heater electrode 122 is arranged below the electrode 121 in the ceramic plate 120, and generates heat when a voltage is applied thereto. Due to heat generated by the heater electrode 122, the ceramic plate 120 heats the ceramic plate 120 and the target object such as a wafer adsorbed on the upper surface 120b of the ceramic plate 120.

The adhesive layer 130 bonds the base plate 110 and the ceramic plate 120. The adhesive layer 130 has a property (hereinafter, appropriately referred to as 'thermal anisotropy') that thermal conductivity in a stack direction in which the base plate 110 and the ceramic plate 120 are stacked (which may also be simply referred to as 'stack direction', hereinafter) is higher than thermal conductivity in a plane direction perpendicular to the stack direction. Specifically, the adhesive layer 130 has a structure in which a plurality of carbon nanotubes 131, each having thermal conductivity in a longitudinal direction higher than thermal conductivity in other directions, are embedded in a resin 132.

The plurality of carbon nanotubes 131 are linear crystals made of carbon, and are arranged adjacent to each other so that the longitudinal direction faces toward the stack direction of the base plate 110 and the ceramic plate 120. In other words, the plurality of carbon nanotubes 131 are arranged adjacent to each other so that the longitudinal direction of each carbon nanotube 131 is along the stack direction of the base plate 110 and the ceramic plate 120. The thermal conductivity of the carbon nanotubes 131 in the longitudinal direction is higher than that of the ceramic plate 120. The carbon nanotube 131 is an example of the heat transfer body.

The resin 132 is filled between the adjacent carbon nanotubes 131 and is bonded to the ceramic plate 120 and the base plate 110. As the resin 132, for example, a thermosetting resin, a thermoplastic resin, or the like can be used. As the thermosetting resin, for example, an epoxy-based resin, a polyimide-based resin, a phenol-based resin, a silicone-based resin or the like can be used. As the thermoplastic resin, for example, a polyethylene-based resin, a polyphenylene-based resin, a polyether-based resin, an acrylic resin, a fluorine-based resin or the like can be used.

By bonding the base plate 110 and the ceramic plate 120 with the adhesive layer 130 having a structure in which the plurality of carbon nanotubes 131 are embedded in the resin 132, the heat can be smoothly transferred through the adhesive layer 130, as compared with a case in which an adhesive is used. For this reason, the heat transfer property from the ceramic plate 120 to the base plate 110 via the adhesive layer 130 is improved, and therefore, the thermal responsiveness in the adhesive layer 130 can be improved. In addition, since the resin 132 is bonded to the ceramic plate 120 and the base plate 110, the bonding strength of the ceramic plate 120 and the base plate 110 is increased, as compared with a case in which an adhesive layer composed of the carbon nanotube assembly is used. Thereby, even when the stress due to a difference in thermal expansion coefficient between the ceramic plate 120 and the base plate 110 acts on the adhesive layer 130, the peeling off of the adhesive layer 130 can be suppressed and the adhesiveness in the adhesive layer 130 can be improved.

In addition, the resin 132 covers the carbon nanotubes 131 in a state in which upper and lower end surfaces of the carbon nanotubes 131 in the longitudinal direction are exposed. Thereby, since the upper end surfaces of the carbon nanotubes 131 in the longitudinal direction can be brought into contact with the ceramic plate 120 and the lower end

surfaces of the carbon nanotubes **131** in the longitudinal direction can be brought into contact with the base plate **110**, the heat can be further smoothly transferred via the adhesive layer **130**.

Next, a manufacturing method of the substrate fixing device **100** configured as described above will be described with reference to FIG. 3. FIG. 3 is a flowchart showing a manufacturing method of the substrate fixing device **100** according to the first embodiment.

First, a ceramic plate **120** for adsorbing a target object such as a wafer is formed (step **S101**). Specifically, a plurality of green sheets made of, for example, aluminum oxide as a main material are fabricated. An electrode **121** is appropriately formed on one surface of the green sheet, and a heater electrode **122** is formed on one surface of another green sheet. The electrode **121** and the heater electrode **122** may be each formed by screen printing a metal paste on the surface of the green sheet, for example. Then, a ceramic plate **120** is formed by stacking and firing the plurality of green sheets. The ceramic plate **120** has a layer of the electrode **121** and a layer of the heater electrode **122** embedded therein, as shown in FIG. 4, for example. FIG. 4 shows a specific example of the ceramic plate **120**. Note that, if necessary, the heater electrode **122** may be omitted.

When the ceramic plate **120** is formed, an adhesive layer is stacked on the ceramic plate **120** (step **S102**). Specifically, for example, as shown in FIG. 5, an adhesive layer **130** composed of a plurality of carbon nanotubes **131** and resin sheets **132a** and **132b** is temporarily bonded to a lower surface **120a** of the ceramic plate **120** from the resin sheet **132a** side. FIG. 5 shows a specific example of an adhesive layer stacking process. The resin sheets **132a** and **132b** are sheet-like members formed using a semi-cured resin for forming the resin **132**, and support a plurality of carbon nanotubes **131** interposed therebetween. The resin sheet **132a** is connected to upper end portions of the carbon nanotubes **131**, and the resin sheet **132b** is connected to lower end portions of the carbon nanotubes **131**. In the step in which the adhesive layer **130** is stacked on the ceramic plate **120**, the resin **132** is not yet filled between the adjacent carbon nanotubes **131** in the adhesive layer **130**.

Then, the ceramic plate **120** having the adhesive layer **130** stacked thereon is stacked on the base plate **110** via the adhesive layer **130** (step **S105**). Specifically, for example, as shown in FIG. 6, the ceramic plate **120** having the adhesive layer **130** stacked thereon is temporarily bonded to the upper surface **110a** of the base plate **110** from the resin sheet **132b** side of the adhesive layer **130**. FIG. 6 shows a specific example of a ceramic plate stacking process.

Then, the ceramic plate **120** stacked on the base plate **110** via the adhesive layer **130** is bonded to the base plate **110** (step **S104**). Specifically, the semi-cured resin forming the resin sheets **132a** and **132b** is filled between the adjacent carbon nanotubes **131** in the adhesive layer **130** by heating and pressurization, and the filled resin is cured with being in contact with the ceramic plate **120** and the base plate **110**. Thereby, the resin **132** filled between the adjacent carbon nanotubes **131** and bonded to the ceramic plate **120** and the base plate **110** is formed. As a result, the base plate **110** and the ceramic plate **120** are bonded by the adhesive layer **130**, so that a substrate fixing device **100** is completed.

In the substrate fixing device **100**, since the base plate **110** and the ceramic plate **120** are bonded by the adhesive layer **130** having a structure in which the plurality of carbon nanotubes **131** are embedded in the resin **132**, the thermal responsiveness and adhesiveness in the adhesive layer **130** can be improved.

As described above, a substrate fixing device (e.g., substrate fixing device **100**) according to the first embodiment includes a base plate (e.g., base plate **110**), a ceramic plate (e.g., ceramic plate **120**), and an adhesive layer (e.g., adhesive layer **130**). The ceramic plate is fixed to the base plate and adsorbs a substrate (e.g., wafer) by electrostatic force. The adhesive layer bonds the base plate and the ceramic plate. The adhesive layer includes a plurality of linear heat transfer bodies (e.g., carbon nanotubes **131**) and a resin (e.g., resin **132**). The plurality of heat transfer bodies are arranged adjacent to each other so that a longitudinal direction faces towards the stack direction of the ceramic plate and the base plate. In other words, the plurality of heat transfer bodies are arranged adjacent to each other so that the longitudinal direction of each heat transfer body is along the stack direction of the ceramic plate and the base plate. The resin is filled between the adjacent heat transfer bodies and is bonded to the ceramic plate and the base plate. Thereby, according to the substrate fixing device of the first embodiment, the thermal responsiveness and adhesiveness in the adhesive layer can be improved.

In addition, the ceramic plate may have an electrode (e.g., heater electrode **122**) embedded therein for generating heat. Thereby, according to the substrate fixing device of the first embodiment, it is possible to adjust a temperature of the ceramic plate and a temperature of the substrate adsorbed on the ceramic plate to desired temperatures.

Second Embodiment

A second embodiment is different from the first embodiment in that the ceramic plate and the base plate are bonded using an adhesive in addition to the adhesive layer.

FIG. 7 is a schematic view showing a cross section of the substrate fixing device **100** according to the second embodiment. In FIG. 7, the same parts as those in FIG. 2 are denoted with the same reference signs. The substrate fixing device **100** shown in FIG. 7 includes a first adhesive **140** and a second adhesive **150**, in addition to each part shown in FIG. 2.

The first adhesive **140** is arranged between the ceramic plate **120** and the adhesive layer **130**. The second adhesive **150** is arranged between the base plate **110** and the adhesive layer **130**. The first adhesive **140** and the second adhesive **150** may be formed of, for example, the same resin as the resin **132** of the adhesive layer **130** or a resin different from the resin **132**. As the resin forming the first adhesive **140** and the second adhesive **150**, for example, a thermosetting resin, a thermoplastic resin or the like can be used. As the thermosetting resin, for example, an epoxy-based resin, a polyimide-based resin, a phenol-based resin, a silicone-based resin or the like can be used. As the thermoplastic resin, for example, a polyethylene-based resin, a polypolyethylene-based resin, a polyether-based resin, an acrylic resin, a fluorine-based resin or the like can be used.

The adhesive layer **130** is bonded to the ceramic plate **120** via the first adhesive **140** and is bonded to the base plate **110** via the second adhesive **150**. In other words, the ceramic plate **120** is bonded to the base plate **110** by the adhesive layer **130**, the first adhesive **140** and the second adhesive **150**. With such a configuration, the bonding strength of the ceramic plate **120** and the base plate **110** is increased, as compared with the case in which the ceramic plate **120** is bonded to the base plate **110** only by the adhesive layer **130**.

Next, a manufacturing method of the substrate fixing device **100** configured as described above will be described with reference to FIG. 8. FIG. 8 is a flowchart showing a

manufacturing method of the substrate fixing device **100** according to the second embodiment. Note that, in FIG. **8**, the same parts as those in FIG. **3** are denoted with the same reference signs.

When the ceramic plate **120** is formed in step **S101**, an adhesive layer is stacked on the ceramic plate **120** (step **S202**). Specifically, for example, as shown in FIG. **9**, the adhesive layer **130** composed of the plurality of carbon nanotubes **131** and the resin sheets **132a** and **132b** is temporarily bonded to the lower surface **120a** of the ceramic plate **120** from the resin sheet **132a** side via the first adhesive **140**. FIG. **9** shows a specific example of an adhesive layer stacking process. In the step in which the adhesive layer **130** is stacked on the ceramic plate **120**, the resin **132** is not yet filled between the adjacent carbon nanotubes **131** in the adhesive layer **130**.

Then, the ceramic plate **120** having the adhesive layer **130** stacked thereon is stacked on the base plate **110** via the first adhesive **140**, the adhesive layer **130**, and the second adhesive **150** (step **S203**). Specifically, for example, as shown in FIG. **10**, the ceramic plate **120** having the adhesive layer **130** stacked thereon is temporarily bonded to the upper surface **110a** of the base plate **110** from the resin sheet **132b** side of the adhesive layer **130** via the second adhesive **150**. FIG. **10** shows a specific example of a ceramic plate stacking process.

Then, the ceramic plate **120** stacked on the base plate **110** via the first adhesive **140**, the adhesive layer **130**, and the second adhesive **150** is bonded to the base plate **110** (step **S204**). Specifically, the semi-cured resin forming the resin sheets **132a** and **132b** is filled between the adjacent carbon nanotubes **131** in the adhesive layer **130** by heating and pressurization, and the filled resin is cured with being in contact with the first adhesive **140** and the second adhesive **150**. Thereby, the resin **132** filled between the adjacent carbon nanotubes **131** is bonded to the ceramic plate **120** via the first adhesive **140** and is bonded to the base plate **110** via the second adhesive **150**. As a result, the base plate **110** and the ceramic plate **120** are bonded by the first adhesive **140**, the adhesive layer **130**, and the second adhesive **150**, so that the substrate fixing device **100** is completed.

As described above, the substrate fixing device according to the second embodiment further includes a first adhesive (e.g., first adhesive **140**) and a second adhesive (e.g., second adhesive **150**). The first adhesive is arranged between the ceramic plate and the adhesive layer. The second adhesive is arranged between the base plate and the adhesive layer. The adhesive layer is bonded to the ceramic plate via the first adhesive, and is bonded to the base plate via the second adhesive. Thereby, according to the substrate fixing device of the second embodiment, the bonding strength of the ceramic plate and the base plate is increased, as compared with the case in which the ceramic plate is bonded to the base plate only by the adhesive layer.

Note that, in the present embodiment, the ceramic plate **120** has the heater electrode **122** embedded therein. However, the arrangement of the heater electrode is not limited thereto and can be variously changed. Below, modified embodiments of the substrate fixing device **100** will be specifically described.

FIG. **11** shows a first modified embodiment of the substrate fixing device **100** according to the second embodiment. In FIG. **11**, the same parts as those in FIG. **7** are denoted with the same reference signs.

In the modified embodiment shown in FIG. **11**, a heater electrode **141** is arranged in the first adhesive **140**. The heater electrode **141** generates heat when a voltage is

applied thereto. Due to the heat generated by the heater electrode **141**, the first adhesive **120** heats the ceramic plate **120** and a target object such as a wafer adsorbed on the upper surface **120b** of the ceramic plate **120**. In this way, the temperature of the ceramic plate **120** and the temperature of the target object such as a wafer adsorbed on the ceramic plate **120** can be adjusted to desired temperatures.

FIG. **12** shows a second modified embodiment of the substrate fixing device **100** according to the second embodiment. In FIG. **12**, the same parts as those in FIG. **7** are denoted with the same reference signs.

In the modified embodiment shown in FIG. **12**, a heater electrode **151** is arranged in the second adhesive **150**. The heater electrode **151** generates heat when a voltage is applied thereto. Due to the heat generated by the heater electrode **141**, the second adhesive **150** heats the ceramic plate **120** and the target object such as a wafer adsorbed on the upper surface **120b** of the ceramic plate **120**. In this way, the temperature of the ceramic plate **120** and the temperature of the target object such as a wafer adsorbed on the ceramic plate **120** can be adjusted to desired temperatures.

Note that the heater electrode may be individually arranged in the ceramic plate **120**, the first adhesive **140**, and the second adhesive **150**, or may be arranged in at least one of the ceramic plate **120**, the first adhesive **140**, or the second adhesive **150**.

Third Embodiment

A third embodiment is different from the first embodiment, in terms of a structure for protecting an outer surface of an adhesive layer against plasma.

FIG. **13** is a schematic view showing a cross section of the substrate fixing device **100** according to the third embodiment. In FIG. **13**, the same parts as those in FIG. **2** are denoted with the same reference signs. In the substrate fixing device **100** shown in FIG. **13**, an outer surface of the adhesive layer **130** is located on a more inner side than an outer surface of the ceramic plate **120**. The outer surface of the adhesive layer **130** forms a concave portion **135** at an outer circumference of the substrate fixing device **100**, together with the lower surface **120a** of the ceramic plate and the upper surface **110a** of the base plate **110** facing each other. A seal member **160** for closing the concave portion **135** is arranged in the concave portion **135**. In this way, the outer surface of the adhesive layer **130** can be protected against plasma.

That is, if the seal member **160** is not arranged, when the substrate fixing device **100** is used for plasma etching of a wafer, for example, the outer surface of the adhesive layer **130** may be exposed to plasma and consumed. In contrast, the seal member **160** is arranged in the concave portion **135** formed by the outer surface of the adhesive layer **130**, and the lower surface **120a** of the ceramic plate and the upper surface **110a** of the base plate **110** facing each other, thereby inhibiting plasma from reaching the outer surface of the adhesive layer **130**. As a result, contact of the plasma with the adhesive layer **130** can be suppressed, and thus consumption of the adhesive layer **130** by the plasma can be suppressed.

Next, a manufacturing method of the substrate fixing device **100** configured as described above will be described with reference to FIG. **14**. FIG. **14** is a flowchart showing a manufacturing method of the substrate fixing device **100** according to the third embodiment. Note that, in FIG. **14**, the same parts as those in FIG. **3** are denoted with the same reference signs.

When the ceramic plate **120** is formed in step **S101**, an adhesive layer is stacked on the ceramic plate **120** (step **S302**). Specifically, for example, as shown in FIG. **15**, the adhesive layer **130** composed of the plurality of carbon nanotubes **131** and the resin sheets **132a** and **132b** is temporarily bonded to the lower surface **120a** of the ceramic plate **120** from the resin sheet **132a** side. FIG. **15** shows a specific example of an adhesive layer stacking process. The outer surface of the adhesive layer **130** is located on a more inner side than the outer surface of the ceramic plate **120**. In the step in which the adhesive layer **130** is stacked on the ceramic plate **120**, the resin **132** is not yet filled between the adjacent carbon nanotubes **131** in the adhesive layer **130**.

Then, the ceramic plate **120** having the adhesive layer **130** stacked thereon is stacked on the base plate **110** via the adhesive layer **130** (step **S303**). Specifically, for example, as shown in FIG. **16**, the ceramic plate **120** having the adhesive layer **130** stacked thereon is temporarily bonded to the upper surface **110a** of the base plate **110** from the resin sheet **132b** side of the adhesive layer **130**. FIG. **16** shows a specific example of a ceramic plate stacking process.

Then, the ceramic plate **120** stacked on the base plate **110** via the adhesive layer **130** is bonded to the base plate **110** (step **S304**). Specifically, the semi-cured resin forming the resin sheets **132a** and **132b** is filled between the adjacent carbon nanotubes **131** in the adhesive layer **130** by heating and pressurization, and the filled resin is cured with being in contact with the first adhesive **140** and the second adhesive **150**. Thereby, the resin **132** filled between the adjacent carbon nanotubes **131** and bonded to the ceramic plate **120** and the base plate **110** is formed. As a result, for example, as shown in FIG. **17**, an intermediate structure of the substrate fixing device **100** in which the base plate **110** and the ceramic plate **120** are bonded by the adhesive layer **130** is obtained. FIG. **17** shows a specific example of an intermediate structure. In such an intermediate structure, the concave portion **135** is formed at the outer circumference of the intermediate structure by the outer surface of the adhesive layer **130**, and the lower surface **120a** of the ceramic plate **120** and the upper surface **110a** of the base plate **110** facing each other.

When the intermediate structure is obtained, the seal member **160** for closing the concave portion **135** is arranged in the concave portion **135** at the outer circumference of the intermediate structure (step **S305**). Specifically, for example, the annular seal member **160** fits into the concave portion **135**. As the annular seal member **160**, for example, an O-ring, a potting resin or the like can be used. As the potting resin, for example, an epoxy-based resin, a silicone-based resin, a urethane-based resin or the like can be used. The seal member **160** is arranged in the concave portion **135**, so that the substrate fixing device **100** is completed.

As described above, in the substrate fixing device according to the third embodiment, the outer surface of the adhesive layer is located on the more inner side than the outer surface of the ceramic plate. The outer surface of the adhesive layer forms a concave portion (e.g., concave portion **135**) at the outer circumference of the substrate fixing device, together with the adhesive surface (e.g., lower surface **120a**) of the ceramic plate and the adhesive surface (e.g., upper surface **110a**) of the base plate facing each other. In the concave portion, a seal member (e.g., seal member **160**) for closing the concave portion is arranged. Thereby, according to the substrate fixing device of the third embodiment, even when the substrate fixing device **100** is used for, for example, plasma etching of a wafer, consumption of the adhesive layer by plasma can be suppressed.

A fourth embodiment relates to a variation in structure of the adhesive layer **130** and the seal member **160** of the third embodiment.

FIG. **18** is a schematic view showing a cross section of the substrate fixing device **100** according to the fourth embodiment. In FIG. **18**, the same parts as those in FIG. **13** are denoted with the same reference signs. In the substrate fixing device **100** shown in FIG. **18**, the adhesive layer **130** has, as the resin **132**, a first resin **132A** and a second resin **132B** separated from each other. The first resin **132A** covers the upper end portions of the carbon nanotubes **131** and is bonded to the ceramic plate **120**. The second resin **132B** covers the lower end portions of the carbon nanotubes **131** and is bonded to the base plate **110**. The seal member **160** maintains a gap between the ceramic plate **120** and the base plate **110** so that a space **136** is formed between the first resin **132A** and the second resin **132B**. That is, the seal member **160** has a thickness that does not allow the first resin **132A** and the second resin **132B** to contact each other. In this way, even when the stress due to the difference in thermal expansion coefficient between the ceramic plate **120** and the base plate **110** acts on the adhesive layer **130**, the stress is absorbed in the space **136** between the first resin **132A** and the second resin **132B**, and therefore, the damage to the adhesive layer **130** can be suppressed.

Next, a manufacturing method of the substrate fixing device **100** configured as described above will be described with reference to FIG. **19**. FIG. **19** is a flowchart showing a manufacturing method of the substrate fixing device **100** according to the fourth embodiment. Note that, in FIG. **19**, the same parts as those in FIG. **14** are denoted with the same reference signs.

In step **S303**, the ceramic plate **120** is stacked on the base plate **110** via the adhesive layer **130**, so that the outer surface of the adhesive layer **130** forms the concave portion **135** together with the lower surface **120a** of the ceramic plate and the upper surface **110a** of the base plate **110** facing each other. The seal member **160** for closing the concave portion **135** is arranged in the concave portion **135** (step **S404**). Specifically, for example, as shown in FIG. **20**, the annular seal member **160** is placed on a bottom surface of the concave portion **135**. FIG. **20** shows a specific example of a seal member arranging process. As the annular seal member **160**, for example, an O-ring, a resin spacer or the like can be used.

When the seal member **160** is arranged, the ceramic plate **120** is bonded to the base plate **110** (step **S405**). Specifically, the semi-cured resin forming the resin sheet **132a** is filled between the upper end portions of the adjacent carbon nanotubes **131** in the adhesive layer **130** by heating and pressurization, and the filled resin is cured with being in contact with the ceramic plate **120**. Thereby, the first resin **132A** filled between the upper end portions of the adjacent carbon nanotubes **131** and bonded to the ceramic plate **120** is formed. In addition, the semi-cured resin forming the resin sheet **132b** is filled between the lower end portions of the adjacent carbon nanotubes **131** in the adhesive layer **130** by heating and pressurization, and the filled resin is cured with being in contact with the base plate **110**. Thereby, the second resin **132B** filled between the lower end portions of the adjacent carbon nanotubes **131** and bonded to the base plate **110** is formed. Upon the formation of the first resin **132A** and the second resin **132B**, a gap between the base plate **110** and the ceramic plate **120** is maintained by the seal member **160** so that a space is formed between the first resin **132A** and the

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second resin 132B. When the gap between the base plate 110 and the ceramic plate 120 reaches a desired gap, the heating and pressurization are released, and thus the substrate fixing device 100 in which the base plate 110 and the ceramic plate 120 are bonded by the adhesive layer 130 is completed.

As described above, in the substrate fixing device according to the fourth embodiment, the adhesive layer includes the first adhesive (e.g., first resin 132A) and the second adhesive (e.g., second resin 132B). The first resin covers one end portions (e.g., upper end portions) of the heat transfer bodies (e.g., carbon nanotubes 131) and is bonded to the ceramic plate. The second resin covers the other end portions (e.g., lower end portions) of the heat transfer bodies and is bonded to the base plate. The seal member maintains the gap between the ceramic plate and the base plate so that a space is formed between the first resin and the second resin. Thereby, according to the substrate fixing device of the fourth embodiment, the damage to the adhesive layer can be suppressed.

This disclosure further encompasses various exemplary embodiments, for example, described below.

[1] A manufacturing method of a substrate fixing device, the manufacturing method comprising:

forming a ceramic plate having an electrode embedded therein for generating heat;

stacking, on the ceramic plate, an adhesive layer comprising a plurality of linear heat transfer bodies arranged adjacent to each other so that a longitudinal direction is along a stack direction of the ceramic plate and a base plate, and a resin sheet connected to end portions of the heat transfer bodies;

stacking the ceramic plate on the base plate via the adhesive layer; and

bonding the ceramic plate to the base plate by filling a resin for forming the resin sheet between adjacent heat transfer bodies of the heat transfer bodies in the adhesive layer by heating and pressurization to bond the resin to the ceramic plate and the base plate.

[2] The manufacturing method according to [1], wherein the stacking of the adhesive layer comprises stacking the adhesive layer on the ceramic plate via a first adhesive,

wherein the stacking of the ceramic plate comprises stacking the ceramic plate on the base plate via the first adhesive, the adhesive layer, and a second adhesive, and

wherein the bonding of the ceramic plate comprises bonding the resin filled between the adjacent heat transfer bodies in the adhesive layer to the ceramic plate via the first adhesive and to the base plate via the second adhesive.

[3] The manufacturing method according to [1], wherein the stacking of the adhesive layer further comprises: stacking, on the ceramic plate, the adhesive layer whose outer surface is located on a more inner side than an outer surface of the ceramic plate, and

arranging a seal member for closing a concave portion in the concave portion formed by the outer surface of the adhesive layer after the bonding of the ceramic plate, and an adhesive surface of the ceramic plate and an adhesive surface of the base plate facing each other.

[4] The manufacturing method according to [1], wherein the stacking of the adhesive layer further comprises: stacking an adhesive layer on the ceramic plate, the adhesive layer comprising a first resin sheet connected to one end portions of the heat transfer bodies and a second resin sheet connected to the other end portions

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of the heat transfer bodies and having an outer surface located on a more inner side than an outer surface of the ceramic plate,

arranging a seal member for closing a concave portion in the concave portion formed by the outer surface of the adhesive layer after the stacking of the ceramic plate, and an adhesive surface of the ceramic plate and an adhesive surface of the base plate facing each other, and wherein the bonding of the ceramic plate comprises:

filling a first resin forming the first resin sheet between one end portions of the adjacent heat transfer bodies in the adhesive layer, and filling a second resin forming the second resin sheet between the other end portions of the adjacent heat transfer bodies by heating and pressurization, and

maintaining a gap between the base plate and the ceramic plate by the seal member so that a space is formed between the first resin and the second resin.

What is claimed is:

1. A substrate fixing device comprising:

a base plate;

a ceramic plate fixed to the base plate, the ceramic plate configured to adsorb a substrate by electrostatic force; and

an adhesive layer bonding the base plate and the ceramic plate,

wherein the adhesive layer comprises:

a plurality of linear heat transfer bodies arranged adjacent to each other such that a longitudinal direction of each linear heat transfer body is along a stack direction of the ceramic plate and the base plate, and a resin filled between adjacent heat transfer bodies of the heat transfer bodies and bonded to the ceramic plate and the base plate,

wherein each of the heat transfer bodies is a carbon nanotube, and

wherein the resin covers the carbon nanotubes in a state in which both end surfaces of each of the carbon nanotubes in the longitudinal direction are exposed.

2. The substrate fixing device according to claim 1, further comprising:

a first adhesive arranged between the ceramic plate and the adhesive layer; and

a second adhesive arranged between the base plate and the adhesive layer,

wherein the adhesive layer is bonded to the ceramic plate via the first adhesive and is bonded to the base plate via the second adhesive.

3. The substrate fixing device according to claim 2, wherein at least one of the ceramic plate, the first adhesive, or the second adhesive has an electrode embedded therein for generating heat.

4. The substrate fixing device according to claim 1, wherein an outer surface of the adhesive layer is located on a more inner side than an outer surface of the ceramic plate and is configured to form a concave portion at an outer circumference of the substrate fixing device, together with an adhesive surface of the ceramic plate and an adhesive surface of the base plate facing each other, and

wherein the substrate fixing device comprise a seal member arranged in the concave portion and closing the concave portion.

5. The substrate fixing device according to claim 4, wherein the resin includes a first resin and a second resin separated from each other,

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wherein the first resin is configured to cover one end portions of the heat transfer bodies and is bonded to the ceramic plate,

wherein the second resin is configured to cover the other end portions of the heat transfer bodies and is bonded 5 to the base plate, and

wherein the seal member is configured to maintain a gap between the ceramic plate and the base plate so that a space is formed between the first resin and the second resin.

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